

[← Go Back](#)

Hardware

Nano ESP32

Tutorials

[Arduino Nano ESP32 User Manual](#)[Arduino Nano ESP32 IoT Cloud Setup Guide](#)[Debugging with the Nano ESP32](#)[Device to Device Communication with ESP-NOW](#)[Getting Started with Nano ESP32](#)[Nano ESP32 Selecting Pin Configuration](#)[SPIFFS Partition on Nano ESP32](#)[Home](#) / [Hardware](#) / [Nano ESP32](#) /

Nano ESP32 Selecting Pin Configuration

Nano ESP32 Selecting Pin Configuration

Learn how to switch between default & ESP32 pin configurations when programming your board.

Author · Karl Söderby · Last revision · 12/18/2023

ON THIS PAGE

Overview

[Compatibility Mode](#)[Software & Hardware Needed](#)[Different Pin Configurations +](#)[Change Pin Configuration](#)[Nano ESP32 Pin Map](#)[Summary](#)

Overview

The [Arduino Nano ESP32](#) is a Nano form factor board based on an ESP32-S3 SoC. This board is part of the [Arduino Nano Family](#), and follows the same pinout as all Nano boards. This is very convenient if you want to port a project from another Nano board, as you can preserve the same wiring and pin numbers in the code.

This means that if you want to control a pin you can simply use the pin number that is printed on the board itself:

```
1 // This will enable the
2 digitalWrite(8, HIGH);
```

[Help](#)

Compatibility Mode

However, some libraries previously written for ESP32 boards (to name a few: OneWire, FastLED, ESP32Servo) don't support the pin numbers printed on the board and require you to refer to the internal microcontroller's GPIO number instead. For instance, to refer to the pin labelled "D8" on the board such libraries expect you to write `17` in your code as that's the GPIO number corresponding to that pin.

For this purpose, we added a compatibility mode so that you can choose which numbering scheme you want to use:

when porting an existing Arduino project from another Nano board, it's convenient to use the "Arduino pin" scheme which reflects the numbers printed on the board

when using third-party libraries for ESP32, you should use the "GPIO number" scheme which corresponds to the internal numbers

To change this configuration, simply connect your board, go to **Tools > Pin Numbering** and then select your option. More details are available in this tutorial.

Software & Hardware Needed

[Arduino Nano ESP32](#)

[Arduino IDE](#)

above). Can be installed directly in the Arduino IDE.

Different Pin Configurations

So why does the Nano ESP32 pins not match the ESP32 (MCU) pins? The Nano ESP32 was designed using the **Nano form factor**, a favorable form factor for many, which has consistently kept its pins for many years.

This makes it possible to migrate an older Nano board, to a newer generation Nano (like the Nano ESP32), without having to change your hardware setup.

This of course brings a separate issue, which is that this numbering does not match the ESP32's native GPIO assignment. An example of this is:

Pin `2` is actually GPIO `5`

Pin `5` is actually GPIO `8`

and so on (see the full [pin map](#)).

It is common in a board's design that the actual microcontroller's pins don't match the header pins.

Default & Legacy Options

Luckily, we have two configurations available to choose between:

By Arduino pin
([default](#))

By GPIO number ([legacy](#))

So, let's say we are configuring a pin in a sketch, using the

Arduino pin (default) option:

```
1 // with default configu
2 pinMode(2, OUTPUT);
```

But, when using the

GPIO number (legacy) option,
we would need to configure it like
this:

```
1 // with ESP32 configura
2 pinMode(5, OUTPUT);
```

Pin Labels

You can also control pins using labels such as D0, D1, D2.

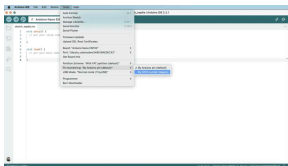
These labels are predefined in the Board Package, and are adjusted based on what configuration you make.

For example, the following code will access the same pin, regardless of what configuration you use:

```
1 /*
2 This will configure the
3 (but will internally co
4 */
5 pinMode(D2, OUTPUT);
```

Change Pin Configuration

To change the pin configuration, open the Arduino IDE, and navigate to **Tools > Pin Numbering**. Here you can select between the default (Nano) and legacy (ESP32) options.



Change pin configuration in the Arduino IDE.

You can now upload a sketch, and the configuration will change.

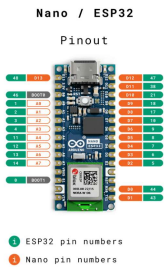
Nano ESP32 Pin Map

To understand how the Nano ESP32 board's pins correlates with the ESP32-S3 SoC pins, have a look at the pin map below:

Nano	ESP32
D0	GPIO44
D1	GPIO43
D2	GPIO5
D3	GPIO6
D4	GPIO7
D5	GPIO8
D6	GPIO9
D7	GPIO10

Nano	ESP32
D9	GPIO18
D10	GPIO21
D11	GPIO38
D12	GPIO47
D13	GPIO48
A0	GPIO1
A1	GPIO2
A2	GPIO3
A3	GPIO4
A4	GPIO11
A5	GPIO12
A6	GPIO13
A7	GPIO14
BOOT0	GPIO46
BOOT1	GPIO0

See the pinout below for a better visual translation:



Nano / ESP32 pinout

Summary

In this tutorial, we've covered how the Nano ESP32's pinout differ from the ESP32-S3 SoC pinout. We've also had a look at how to change the configuration, and provided a pin map that can be

used as a reference when making your next project.

Suggest changes	Need support?	License
The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything wrong, you can edit this page here .	Help Center Ask the Arduino Forum Discover Arduino Discord	The Arduino documentation is licensed under the Creative Commons Attribution-Share Alike 4.0 license.

Was this article helpful?

